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B. Tech. III - Sem. (Main) Exam., (Academic Session 2021- 2022)

Computer Science and Engineering

3CS4 – 06 Object Oriented Programming

Common for CSE, T, CSE(AI&ML), S(DS), CSE(AI),AI&DS, AI&ML

Time: 2½ Hours **Maximum Marks: 120**

Min. Passing Marks:

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 6 × 3 marks = 18 marks. Candidates have to answer* ***six*** *questions out of* ***ten***

Part – B: *Analytical/Problem solving questions 3 × 10 marks = 30 marks. Candidates have to answer* ***three*** *questions out of* ***seven****.*

Part – C: *Descriptive/Analytical/Problem Solving questions 3 ×24 marks = 72 marks. Candidates have to answer* ***three*** *questions out of* ***five****.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No. 205)

1. NIL 2. NIL

PART – A

Q.1 Explain structure as user defined data type.

Q.2 What do you understand by an abstract class? Explain.

Q.3 What is the difference between static and dynamic memory allocation?

Q.4 What do you mean by this pointer? Explain using an example.

Q.5 Differentiate between function overloading and overriding.

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Q.6 Discuss on various access specifiers.

Q.7 Explain difference between static binding and dynamic binding.

Q.8 What is a pure virtual function? Explain.

Q.9 What is generic programming? Explain.

Q.10 Describe the functionality of -

(i) of stream

(ii) if stream

(iii) f stream

PART – B

Q.1 Write a program to pass the structure to a function. Explain each and every step in detail.

Q.2 What is friend function? Explain characteristics of friend function using an example.

Q.3 Define super and sub class. Explain the concept of public, private and protected inheritance. https://www.btubikaner.com

Q.4 What is dynamic binding? Explain the concept of virtual function with the help of an example.

Q.5 Explain try, throw, and catch constructs with their syntax. Also, write a program to implement it.

Q.6 Give the description of the following stream mode flags –

(i) std :: ios :: app (ii) std :: ios :: ate

(iii) std :: ios :: binary (iv) std :: ios :: noreplace

Q.7 Describe the mechanism of accessing data members and member functions in the following case –

(i) Inside the main program

(ii) Inside the member function of same class

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PART – C

Q.1 What do you understand by Object oriented programming? What are it's advantages over procedure oriented programming? Explain the features of object oriented programming paradigm.

Q.2 What is a constructor? List various types of constructor and describe with the help of suitable example.

Q.3 What is inheritance? What are the various forms of inheritance? Explain each with the help of suitable example.

Q.4 What is operator over loading? How will you overload binary and unary operators? Discuss both process with the help of programming implementation.

Q.5 What is template class and template function? Write a function template for sorting an array.

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